

Atlas Embeddings: Exceptional Lie Groups from First Principles Mathematical Foundations and Main Theorems

UOR Foundation

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Abstract

This work establishes a novel connection between combinatorial structures and exceptional Lie theory. We define the **Atlas of Resonance Classes**, a 96-vertex graph arising from variational principles on cell complexes, and prove its canonical embedding into the E_8 root system. Through categorical operations in the category **ResGraph** of resonance graphs, we construct all five exceptional Lie groups (G_2 , F_4 , E_6 , E_7 , E_8) from this single universal object. The central mathematical contribution is the proof that the Atlas embeds uniquely into E_8 and serves as the initial object from which exceptional group structures emerge naturally.

1 Introduction

The classification of simple Lie algebras reveals five exceptional cases: G_2 , F_4 , E_6 , E_7 , and E_8 . While these have been extensively studied since their discovery by Killing and Cartan, their emergence from first principles remains mysterious. This work presents a unified construction from a single combinatorial object: the **Atlas of Resonance Classes**.

2 The E_8 Root System

Definition 1 (E_8 Root System). *The E_8 root system $R_{E_8} \subset \mathbb{R}^8$ consists of 240 vectors of squared norm 2:*

- 112 integer roots: $\pm e_i \pm e_j$ for $1 \leq i < j \leq 8$
- 128 half-integer roots: $(\pm \frac{1}{2}, \pm \frac{1}{2}, \dots, \pm \frac{1}{2})$ with an even number of minus signs

The inner product between distinct roots takes values in $\{0, \pm 1\}$.

Definition 2 (Weyl Group). *The Weyl group $W(E_8)$ is generated by reflections $s_\alpha(x) = x - \frac{2\langle x, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha$ for $\alpha \in R_{E_8}$.*

3 The Atlas of Resonance Classes

Definition 3 (Atlas Graph). *The **Atlas** $\mathcal{A} = (V_{\mathcal{A}}, E_{\mathcal{A}}, \lambda_{\mathcal{A}}, U_{\mathcal{A}})$ is a 96-vertex graph with:*

- $V_{\mathcal{A}}$: 96 vertices representing resonance classes
- $E_{\mathcal{A}} \subseteq V_{\mathcal{A}} \times V_{\mathcal{A}}$: edges defined by the adjacency predicate
- $\lambda_{\mathcal{A}} : V_{\mathcal{A}} \rightarrow R_{E_8}$: vertex labeling into E_8 roots
- $U_{\mathcal{A}} \subseteq V_{\mathcal{A}}$: distinguished unity vertices

Definition 4 (Adjacency Predicate). *For $t \in \{0, \pm 1\}$, define:*

$$\Phi(t) \iff (t = -1)$$

Two vertices $v, w \in V_{\mathcal{A}}$ are adjacent iff $\langle \lambda_{\mathcal{A}}(v), \lambda_{\mathcal{A}}(w) \rangle = -1$.

3.1 The Category of Resonance Graphs

Definition 5 (Resonance Graph Object). *A **ResGraph object** is a tuple $\mathcal{G} = (V, E, \lambda, U)$ where:*

1. V is a finite set of vertices
2. $E \subseteq \{\{v, w\} : v, w \in V, v \neq w\}$ is a set of edges

3. $\lambda : V \rightarrow R_{E_8}$ labels vertices with E_8 roots
4. $U \subseteq V$ is a distinguished unity subset
5. Adjacency rule: $\{v, w\} \in E$ iff $\Phi(\langle \lambda(v), \lambda(w) \rangle)$

Definition 6 (ResGraph Morphism). A morphism $\varphi : (V, E, \lambda, U) \rightarrow (V', E', \lambda', U')$ is a function $\varphi : V \rightarrow V'$ such that:

1. (Graph homomorphism) $\{v, w\} \in E \Rightarrow \{\varphi(v), \varphi(w)\} \in E'$
2. (Label compatibility) $\exists g \in W(E_8)$ such that $\lambda'(\varphi(v)) = g \cdot \lambda(v)$ for all $v \in V$
3. (Unity preservation) $\varphi(U) \subseteq U'$

3.2 Main Theorem: Atlas to E_8 Embedding

Theorem 7 (Atlas Embedding). There exists an injective map $f : V_{\mathcal{A}} \rightarrow R_{E_8}$ such that:

1. For all $v, w \in V_{\mathcal{A}}$, $\{v, w\} \in E_{\mathcal{A}}$ iff $\langle f(v), f(w) \rangle = -1$
2. $|\text{im}(f)| = 96$
3. (Uniqueness up to Weyl group) If f' also satisfies (1) and (2), then there exists $w \in W(E_8)$ with $f' = w \circ f$

Proof Outline. **Existence:** Construct f explicitly by providing coordinates for all 96 vertices in the canonical E_8 basis. Verify adjacency by checking all $\binom{96}{2}$ pairs satisfy Φ .

Uniqueness:

1. Fix a seed vertex v_0 and normalize $f(v_0)$ to a chosen root via Weyl action
2. Neighbors of v_0 must map to the 56 roots with inner product -1 with $f(v_0)$
3. Local adjacency patterns force unique assignments via connectedness of $\mathcal{A}$
4. Any alternative embedding differs by a global Weyl transformation

□

3.3 Categorical Operations in ResGraph

Definition 8 (Reflection Closure). For $\mathcal{G} = (V, E, \lambda, U)$, define $\text{RC}(\mathcal{G})$ as the smallest ResGraph containing $\mathcal{G}$ whose label set is closed under reflections:

$$\Lambda_{k+1} = \Lambda_k \cup \{s_\alpha(\beta) : \alpha, \beta \in \Lambda_k\} \cap R_{E_8}$$

with $\Lambda_0 = \lambda(V)$.

Definition 9 (Resonance Product). For $\mathcal{G}_1, \mathcal{G}_2$, define the partial sum graph:

$$\mathcal{G}_1 \boxtimes \mathcal{G}_2 = \{(v_1, v_2) : \lambda_1(v_1) + \lambda_2(v_2) \in R_{E_8}\} / \sim$$

where $(v_1, v_2) \sim (v'_1, v'_2)$ if their label sums coincide.

Definition 10 (Quotient by Symmetry). For $H \leq \text{Aut}(\mathcal{G})$, define $\mathcal{G}/H$ with vertices as H -orbits, labels by representatives, and edges between orbits when any representatives are adjacent.

3.4 Exceptional Group Constructions

Theorem 11 (G_2 Construction). The exceptional group G_2 emerges as:

$$G_2 = \text{RC}(\mathcal{A} \boxtimes C_3)/H$$

where C_3 is a 3-element resonance graph and H an appropriate symmetry group. The construction yields:

- 12 roots (6 short, 6 long)
- Rank 2, Cartan matrix $\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$
- Weyl group order 12

Theorem 12 (F_4 Construction).

$$F_4 = \text{RC}(\mathcal{A}/\tau)$$

where τ is the mirror symmetry involution, yielding:

- 48 roots (24 short, 24 long)

- Rank 4
- Weyl group order 1152

Theorem 13 (E_6, E_7, E_8 Constructions). • $E_6 = \text{RC}(\mathcal{A}[\ell = c])$ via hyperplane filtration

- $E_7 = \text{Aug}(\mathcal{A})$ via root addition closure
- $E_8 = \text{RC}(\mathcal{A})$ via full reflection closure

3.5 Initiality and Universal Property

Conjecture 14 (Atlas Initiality). *The Atlas $\mathcal{A}$ is initial in **ResGraph**: for every object $\mathcal{G} \in \mathbf{ResGraph}$, there exists a unique morphism $\varphi : \mathcal{A} \rightarrow \mathcal{G}$.*

Definition 15 (Precursor Category). *Define $\mathbf{ResGraph}_0$ with objects (V, E, U) (forgetting labels) and morphisms preserving graph structure and unity. There is a faithful functor $J : \mathbf{ResGraph} \rightarrow \mathbf{ResGraph}_0$.*

Lemma 16 (Lifting Lemma). *Any $\mathbf{ResGraph}_0$ -morphism $f : J(\mathcal{A}) \rightarrow J(\mathcal{G})$ lifts uniquely to a $\mathbf{ResGraph}$ -morphism $\tilde{f} : \mathcal{A} \rightarrow \mathcal{G}$ if $\mathcal{G}$ satisfies the E_8 -liftability condition.*

3.6 Action Functional and Variational Foundation

Definition 17 (Flag-Resonance Complex). *For $\mathcal{G} = (V, E, \lambda, U)$, define the cellular complex $\text{FR}(\mathcal{G})$ by:*

- 0-cells: V
- 1-cells: E
- k -cells: $\text{clique}_{k+1}(\mathcal{G})$ (cliques of size $k + 1$)

Definition 18 (Action Functional). *For $\varphi \in C^{k-1}(\text{FR}(\mathcal{G}); A)$, define:*

$$S_k[\varphi] = \sum_{\sigma \in \text{FR}^{(k)}} \varphi(\partial\sigma)$$

The Atlas emerges as stationary configuration of S_k for appropriate k and coefficient group A .

3.7 Computational Verification Framework

Theorem 19 (Exact Arithmetic Verification). *All constructions and verifications use exact rational arithmetic:*

1. *Vertex coordinates in $\mathbb{Q}$ or $\frac{1}{2}\mathbb{Z}$*
2. *Inner products computed exactly via $\langle x, y \rangle = \sum x_i y_i$*
3. *Adjacency verified by $\Phi(\langle \lambda(v), \lambda(w) \rangle)$*
4. *Cartan matrices derived from simple systems*

Proposition 20 (Reproducible Artifacts). *The theory is computationally certified by:*

- *Complete vertex/edge listings for $\mathcal{A}$*
- *Explicit coordinate tables for all exceptional groups*
- *Verification scripts confirming all mathematical claims*
- *Flag complex cell counts justifying the 12,288-cell structure*